

```
srunk --cpu-freq=2400000-2400000 numactl --membind=0 likwid-bench -t daxpy -w M0:5000MBW
```

```
Time: 6.550983e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6105.95
Data volume (Byte): 479999913984
MByte/s: 73271.43
Cycles per update: 0.784258
Cycles per cacheline: 6.274064
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=0 likwid-bench -t daxpy -w M1:5000MB
```

```
Time: 6.527574e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6127.85
Data volume (Byte): 479999913984
MByte/s: 73534.19
Cycles per update: 0.781453
Cycles per cacheline: 6.251621
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=0 likwid-bench -t daxpy -w M2:5000MB
```

```
Time: 9.057499e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4416.23
Data volume (Byte): 479999913984
MByte/s: 52994.75
Cycles per update: 1.084326
Cycles per cacheline: 8.674611
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
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```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=0 likwid-bench -t daxpy -w M3:5000MB
```

```
Time: 9.034977e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4427.24
Data volume (Byte): 479999913984
MByte/s: 53126.86
Cycles per update: 1.081630
Cycles per cacheline: 8.653043
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
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```

srunk --cpu-freq=2400000-2400000 numactl --membind=1 likwid-bench -t daxpy -w M0:5000MB

```
Time: 6.564544e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6093.34
Data volume (Byte): 479999913984
MByte/s: 73120.07
Cycles per update: 0.785874
Cycles per cacheline: 6.286994
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

srunk --cpu-freq=2400000-2400000 numactl --membind=1 likwid-bench -t daxpy -w M1:5000MB

```
Time: 6.564229e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6093.63
Data volume (Byte): 479999913984
MByte/s: 73123.58
Cycles per update: 0.785842
Cycles per cacheline: 6.286735
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=1 likwid-bench -t daxpy -w M2:5000MB
```

```
Time: 9.087664e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4401.57
Data volume (Byte): 479999913984
MByte/s: 52818.85
Cycles per update: 1.087936
Cycles per cacheline: 8.703492
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=1 likwid-bench -t daxpy -w M3:5000MB
```

```
Time: 9.066638e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4411.78
Data volume (Byte): 479999913984
MByte/s: 52941.33
Cycles per update: 1.085419
Cycles per cacheline: 8.683355
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

srunk --cpu-freq=2400000-2400000 numactl --membind=2 likwid-bench -t daxpy -w M0:5000MB

```
Time: 9.071404e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4409.46
Data volume (Byte): 479999913984
MByte/s: 52913.52
Cycles per update: 1.085978
Cycles per cacheline: 8.687826
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

srunk --cpu-freq=2400000-2400000 numactl --membind=2 likwid-bench -t daxpy -w M1:5000MB

```
Time: 9.075295e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4407.57
Data volume (Byte): 479999913984
MByte/s: 52890.83
Cycles per update: 1.086452
Cycles per cacheline: 8.691618
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=2 likwid-bench -t daxpy -w M2:5000MB
```

```
Time: 9.075295e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4407.57
Data volume (Byte): 479999913984
MByte/s: 52890.83
Cycles per update: 1.086452
Cycles per cacheline: 8.691618
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=2 likwid-bench -t daxpy -w M3:5000MB
```

```
Time: 6.547561e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6109.14
Data volume (Byte): 479999913984
MByte/s: 73309.73
Cycles per update: 0.783847
Cycles per cacheline: 6.270778
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=3 likwid-bench -t daxpy -w M0:5000MB
```

```
Time: 9.077229e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4406.63
Data volume (Byte): 479999913984
MByte/s: 52879.56
Cycles per update: 1.086687
Cycles per cacheline: 8.693495
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=3 likwid-bench -t daxpy -w M1:5000MB
```

```
Time: 9.085849e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 4402.45
Data volume (Byte): 479999913984
MByte/s: 52829.39
Cycles per update: 1.087720
Cycles per cacheline: 8.701757
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=3 likwid-bench -t daxpy -w M2:5000MB
```

```
Time: 6.567773e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6090.34
Data volume (Byte): 479999913984
MByte/s: 73084.12
Cycles per update: 0.786267
Cycles per cacheline: 6.290138
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```

```
srunk --cpu-freq=2400000-2400000 numactl --membind=3 likwid-bench -t daxpy -w M3:5000MB
```

```
Time: 6.553160e+00 sec
Iterations: 1152
Iterations per thread: 64
Inner loop executions: 4340277
Size (Byte): 4999999104
Size per thread: 277777728
Number of Flops: 39999992832
MFlops/s: 6103.92
Data volume (Byte): 479999913984
MByte/s: 73247.10
Cycles per update: 0.784516
Cycles per cacheline: 6.276128
Loads per update: 2
Stores per update: 1
Load bytes per element: 16
Store bytes per elem.: 8
Load/store ratio: 2.00
Instructions: 94999982993
UOPs: 129999976704
```